

```
Script started on 2026-02-10 20:48:34-06:00 [TERM="xterm" TTY="/dev/pts/39" COLUMNS=80]
rc47379@ares:~$ pwd
```

```
/home/students/rc47379
rc47379@ares:~$ cat citydistance.info
```

```
NAME: Calvin Rice
```

```
CLASS: CSC122-002
```

```
TITLE: Lets map it out
```

```
LEVEL: 4
```

```
DESCRIPTION:
```

```
rc47379@ares:~$ show-code citydistance.info
```

```
Unknown class/file type! Please have your teacher request a group update...
rc47379@ares:~$ show-code citydistance.cpp
```

```
citydistance.cpp:
```

```
1  #include <iostream>
2  #include <vector>
3
4  #include "Point.h"
5  #include "City.h"
6  #include "inputprotection.h"
7
8  using namespace std;
9
10 int main()
11 {
12     vector<City> cityList = {};
13     bool choosing = true;
14
15     while(choosing)
16     {
17         cout << "\n1) Enter city Information\n"
18              << "2) Calculate Distance between two cities\n"
19              << "3) Print All cities\n"
20              << "4) Quit" << endl;
21
22         char choice = input_protect("Choose an option: ", "Invalid Choice,
23
24         double xPos, yPos;
25         string _name;
26         bool invalidName = true;
27         City city1, city2;
28         long cityIndex1, cityIndex2;
```

```
29         Point _location;
30
31         cout << endl;
32
33         switch(choice)
34         {
35             case '1':
36                 if(static_cast<int>(cityList.size()) >= 10)
37                 {
38                     cout << "City list is full." << endl;
39                     break;
40                 }
41
42                 while(invalidName)
43                 {
44                     cout << "Enter City Name (No Spaces): ";
45                     cin.clear();
46                     cin >> _name;
47                     cout << endl;
48                     if(_name[0] == '\0')
49                     {
50                         cout << "Invalid Name, Try Again" << endl;
51                     }
52                     else
53                     {
54                         invalidName = false;
55                     }
56                 }
57
58                 cin.clear();
59                 xPos = input_protect("Enter City X Position: ", "Invalid P
60                 yPos = input_protect("Enter City Y Position: ", "Invalid P
61
62                 _location =* new Point(xPos, yPos);
63
64                 cityList.push_back(City(_name, _location));
65                 break;
66             case '2':
67                 if(static_cast<long>(cityList.size()) < 2)
68                 {
69                     cout << "Not Enough Cities Saved." << endl;
70                     break;
71                 }
72
73                 for(int i = 0; i < cityList.size(); i++)
74                 {
75                     cout << i + 1 << ": " << cityList[i].get_name() << endl;
76                 }
77
78                 cin.clear();
79                 cityIndex1 = input_protect("Choose First City: ", "Invalid
80                 cityIndex2 = input_protect("Choose Second City: ", "Invalid
81
82                 city1 = cityList[cityIndex1 - 1];
```

```

83         city2 = cityList[cityIndex2 - 1];
84
85         cout << "The distance between " << city1.get_name() << " and " << city2.get_name() << endl;
86
87         break;
88     case '3':
89         if(static_cast<long>(cityList.size()) == 0)
90         {
91             cout << "No Cities Saved." << endl;
92             break;
93         }
94
95         for(int i = 0; i < static_cast<int>(cityList.size()); i++)
96         {
97             cout << i + 1 << ": " << cityList[i].get_name() << endl;
98         }
99         break;
100    case '4':
101        return 0;
102        break;
103    default:
104        cout << "Enter a valid choice" << endl;
105        break;
106    }
107
108    choice = 0;
109 }
110
111
112     return 0;
113 }

```

rc47379@ares:~\$ CPP citydistance

citydistance.cpp***

citydistance.cpp: In function 'int main()':

citydistance.cpp:62:51: warning: implicitly-declared 'constexpr Point& Point::operator=(const Point&)' is deprecated [-Wdeprecated-copy]

```

62 |         _location =* new Point(xPos,
    |         ^
    |         |
    |         ~~~~~

```

In file included from **citydistance.cpp:4:**

Point.h:13:5: note: because 'Point' has user-provided 'Point::Point(const Point&)'

```

13 |         Point(const Point & p);
    |         ~~~~~

```

citydistance.cpp:73:34: warning: comparison of integer expressions of different signedness: 'int' and 'std::vector<City>::size_type' {aka 'long

unsigned int'} [-Wsign-compare]

```

73 |         for(int i = 0; i <
    |         ~~~~~
    |         |
    |         ~~~~~

```

In file included from **citydistance.cpp:5:**

City.h: In member function 'constexpr City& City::operator=(const City&)':

City.h:8:7: warning: implicitly-declared 'constexpr Point& Point::operator=(const Point&)' is deprecated [-Wdeprecated-copy]

```

8 |     class City
    |     ~~~~~

```

Point.h:13:5: note:

because 'Point' has user-provided 'Point::Point(const Point&)'

```

13 |         Point(const Point & p);
    |         ~~~~~

```

citydistance.cpp: In function 'int main()':

citydistance.cpp:82:48: note: synthesized method 'constexpr City& City::operator=(const City&)' first required here

```

82 |         city1 = cityList[cityIndex1 -
    |         ~~~~~
    |         |
    |         ~~~~~

```

citydistance.cpp:(.text+0x113): undefined reference to `input_protect(std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >)'

citydistance.cpp:(.text._ZN4CityC2E0S_[_ZN4CityC5E0S_]+0x23): undefined reference to `Point::Point(Point const&)'

citydistance.cpp:(.text._ZN4CityC2ERKS_[_ZN4CityC5ERKS_]+0x23): undefined reference to `Point::Point(Point const&)'

collect2: error: ld returned 1 exit status

rc47379@ares:~\$./citydistance.out

bash: ./citydistance.out: No such file or directory

rc47379@ares:~\$./citydistance

```

1) Enter city Information
2) Calculate Distance between two cities
3) Print All cities
4) Quit
Choose an option:
1

```

Enter City Name (No Spaces): newyork

Enter City X Position:

12

Enter City Y Position:

34

- 1) Enter city Information
 - 2) Calculate Distance between two cities
 - 3) Print All cities
 - 4) Quit
- Choose an option:
1

Enter City Name (No Spaces): losangeles

Enter City X Position:

76

Enter City Y Position:

88

- 1) Enter city Information
 - 2) Calculate Distance between two cities
 - 3) Print All cities
 - 4) Quit
- Choose an option:
2

- 1: newyork
 - 2: losangeles
- Choose First City:
1

- Choose Second City:
2

The distance between newyork and losangeles is 83.7377

- 1) Enter city Information
 - 2) Calculate Distance between two cities
 - 3) Print All cities
 - 4) Quit
- Choose an option:
3

- 1: newyork
- 2: losangeles

- 1) Enter city Information
 - 2) Calculate Distance between two cities
 - 3) Print All cities
 - 4) Quit
- Choose an option:
4

rc47379@ares:~\$ cat citydistance.tpq

cat: citydistance.tpq: No such file or directory
rc47379@ares:~\$ exit

exit

Script done on 2026-02-10 20:49:59-06:00 [COMMAND_EXIT_CODE="1"]